AI assistance disclosure

OpenAI Codex, with GPT-6 Sol, GPT-6 Astra, and GPT-6.1 Sol, assisted with proof development, mathematical and scientific checks, drafting, and revision. OpenAI assistance with the synthetic implementation check covered code design, execution, verification, and results wording. The affected text includes the abstract, impact statement, all main sections, and all supplement sections. ChatGPT suggested the sample-mean comparison and the two-example illustration. Anthropic Claude Opus 5.5 also provided scientific assistance with the mathematical arguments and manuscript. The assistance included mathematical content and was not limited to language editing. The author is responsible for the claims and text.